Term 3 answer key

Formulas from memory. Terms 3–4 add the Year 8 families (Pythagoras; trapezium, rhombus and kite, circle area; index laws; percentage change, profit and loss). Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes + Pythagoras	Q2 B Measurement	Q3 C Number & percentages + index laws	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	65°	120 m^3	0.75	$6x + 12$	7/8
	2	143°	452.4 cm^2	$17	4 h 45 min	9/25
	3	102°	60 cm^2	49	$x = 13$	6, 2
2	1	66°	40 cm	55%	2 : 5	13
	2	25 cm	36 cm^2	28 L	50 km/h	1/2
	3	72°	70 cm^2	0.35	32	4
3	1	49°	70 m^2	−1	$150 and $90	7/9
	2	24 cm	33 cm^2	$936	$4x − 12$	9/25
	3	70°	50.3 cm^2	y^5	12 km per litre	2
4	1	116°	31.4 cm	9 m	$x = 16$	(2, 5)
	2	38°	30 cm^2	250	00:30	22 cm
	3	137°	5 cm	18	14	4
5	1	29 cm	66 cm^2	10	$10a − 8$	3/4
	2	72°	54 cm	26%	2 : 5 : 3	1/5
	3	4 m	40 cm^2	25% profit	240 km	21/50
6	1	85°	12 m	35%	$x = 15$	7
	2	100°	153.9 m^2	$36	$20 and $70	(3, 5)
	3	115°	6 cm	6	26	12 °C
7	1	52°	20 m^2	2^7	Box B ($23 per kg)	2/9
	2	34 cm	69.1 cm	63 kg	$2x + 14$	24
	3	81°	21 cm^2	16	23:05	1/4
8	1	12 m	72 m^3	3.8	$x = 7$	1/2
	2	99°	36 cm	$90	11	20
	3	60°	113.1 cm^2	25%	4 : 3	8, 12
9	1	115°	65 m^2	$30	90 km/h	23
	2	142°	30 m^2	12/25	$60 and $90	1/5
	3	127°	56 m^2	24	$14 + 7k$	1/5
10	1	5 m	36 cm^2	7^{10}	$p = 6$	8
	2	20 cm	30 cm^2	−4	34	5/12
	3	65°	25.1 cm	25% loss	$2 per kg	6

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

A MA4-PYT-C-01 Pythagoras' theorem (hypotenuse and shorter side; Pythagorean triples)

B MA4-ARE-C-01 area of trapeziums, rhombuses and kites, and circles (this replaces rectangles and unit conversions in Terms 3–4)

C MA4-IND-C-01 index laws (multiply, divide, power of a power) with numerical bases; the questions with pronumeral bases are Stage 5 content (MA5-IND-C-01), included as asked; MA4-FRC-C-01 percentage increase and decrease, percentage profit and loss*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.